

- 1 (a) work done in bringing unit mass from infinity (to the point) B1 [1]
- (b) gravitational force is (always) attractive B1
 either as r decreases, object/mass/body does work
 or work is done by masses as they come together B1 [2]
- (c) either force on mass = mg (where g is the acceleration of free fall /gravitational field strength) B1
 $g = GM/r^2$ B1
 if $r \approx h$, g is constant B1
 $\Delta E_p = \text{force} \times \text{distance moved}$ M1
 $= mgh$ A0
 or $\Delta E_p = m\Delta\phi$ (C1)
 $= GMm(1/r_1 - 1/r_2) = GMm(r_2 - r_1)/r_1r_2$ (B1)
 if $r_2 \approx r_1$, then $(r_2 - r_1) = h$ and $r_1r_2 = r^2$ (B1)
 $g = GM/r^2$ (B1)
 $\Delta E_p = mgh$ (A0) [4]
- (d) $\frac{1}{2}mv^2 = m\Delta\phi$
 $v^2 = 2 \times GM/r$ C1
 $= (2 \times 4.3 \times 10^{13}) / (3.4 \times 10^6)$ C1
 $v = 5.0 \times 10^3 \text{ m s}^{-1}$ A1 [3]
 (Use of diameter instead of radius to give $v = 3.6 \times 10^3 \text{ m s}^{-1}$ scores 2 marks)

- 2 (a) (i) either random motion